

Security Games in Online Advertising: Can Ads Help Secure the Web?



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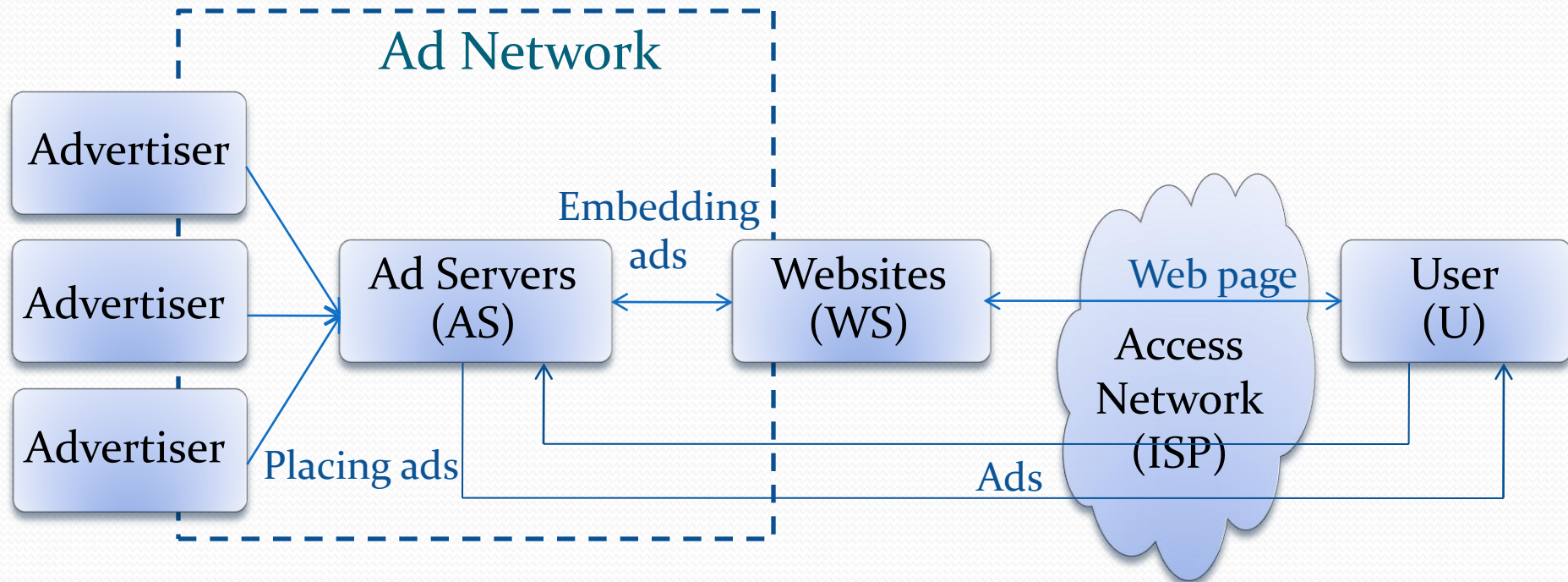
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Internet Economy

- Online Advertising:
 - The main Internet business model
 - Revenue in 2009 in the US is \$22.4 billion
 - Sponsors free services and applications
- What happens if one meddles with it?

Online Advertising System



Role of ISPs

- Traditional role:
 - Provide Internet access to end users
 - Forward the communication in compliance with Network Neutrality Policy
- New requirements
 - Data retention legislations
 - Increase costs and require investing into new technologies
- How will ISPs obtain a return on investment?

Recently Reported Cases



- Growing number of [ISPs injecting own content](#) into web pages [1][2]
- Third party ad companies [partnering with ISPs](#)
 - e.g., Adzilla, Phorm, NebuAd

[1] C. Reis et al. **Detecting In-flight Page Changes with Web Tripwires**, NSDI 2008.

[2] B. April, F. Hacquebord and R. Link, **A Cybercrime Hub**, August 2009.

ISPs in Online Advertising Business

- *Non-cooperative ISP* – diverts part of online ad revenue by performing attacks on online advertising
 - E.g., injecting ads into the content of web pages on-the-fly
- *Cooperative ISP* – collects and provides information about users' online behavior with the goal of improving ad targeting
 - Generates revenue by charging for users' profiles

Problem Statement

- Study the effect of strategic ISPs on the Web
 - Model the behavior of ISPs and economic incentives in online advertising systems
 - Analyze mutually dependent actions of ISPs and Ad Servers (AS)

Related Work

- Online advertising fraud
 - The best strategy for ad networks is to fight click fraud [1]
- Incentives to increase the security of the Web
 - Users' choice: Investment in security or insurance mechanisms [2]
- Our model introduces a new strategic player – the ISP

[1] B. Mungamuru, S. Weis, H. Garcia-Molina, **Should Ad Networks Bother Fighting Click Fraud? (Yes, they should.)**, Stanford Technical Report, July 2008.

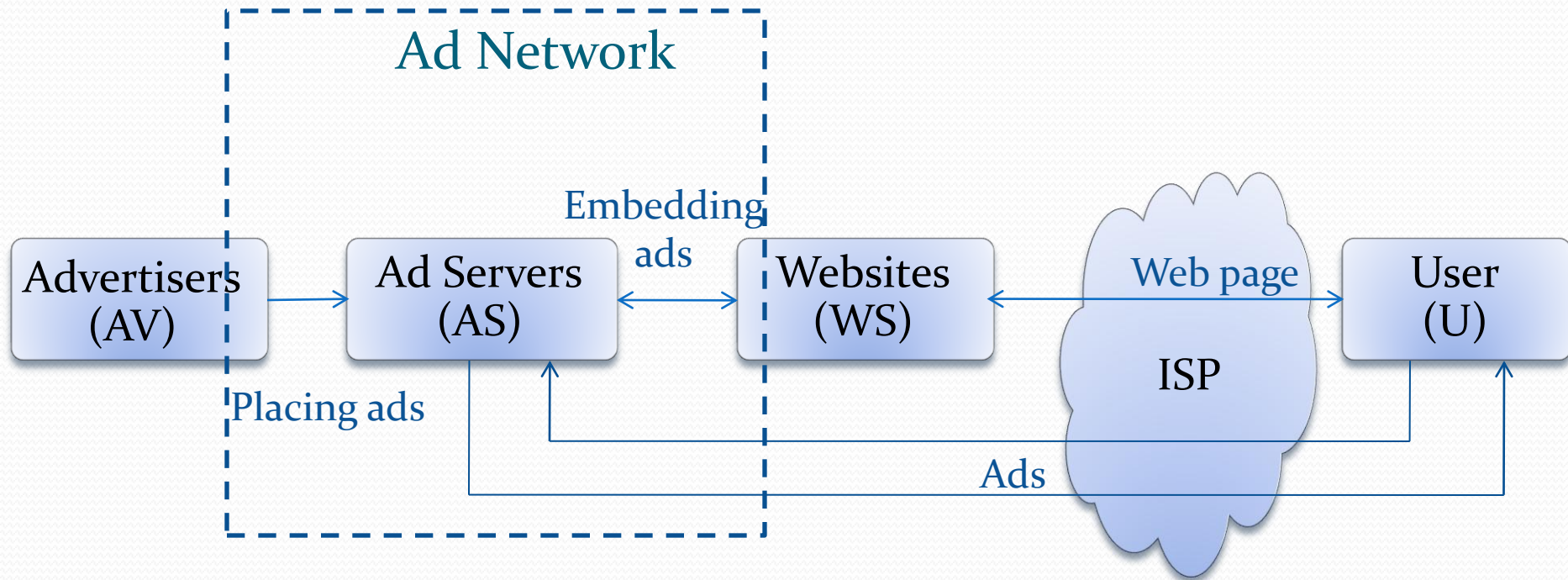
[2] J. Grossklags, N. Christin, J. Chuang, **Secure or insure?: a game-theoretic analysis of information security games**, WWW 2008.

Outline

- I. Strategic behavior of ISPs
- II. Game-theoretic Model
- III. Analysis and Results

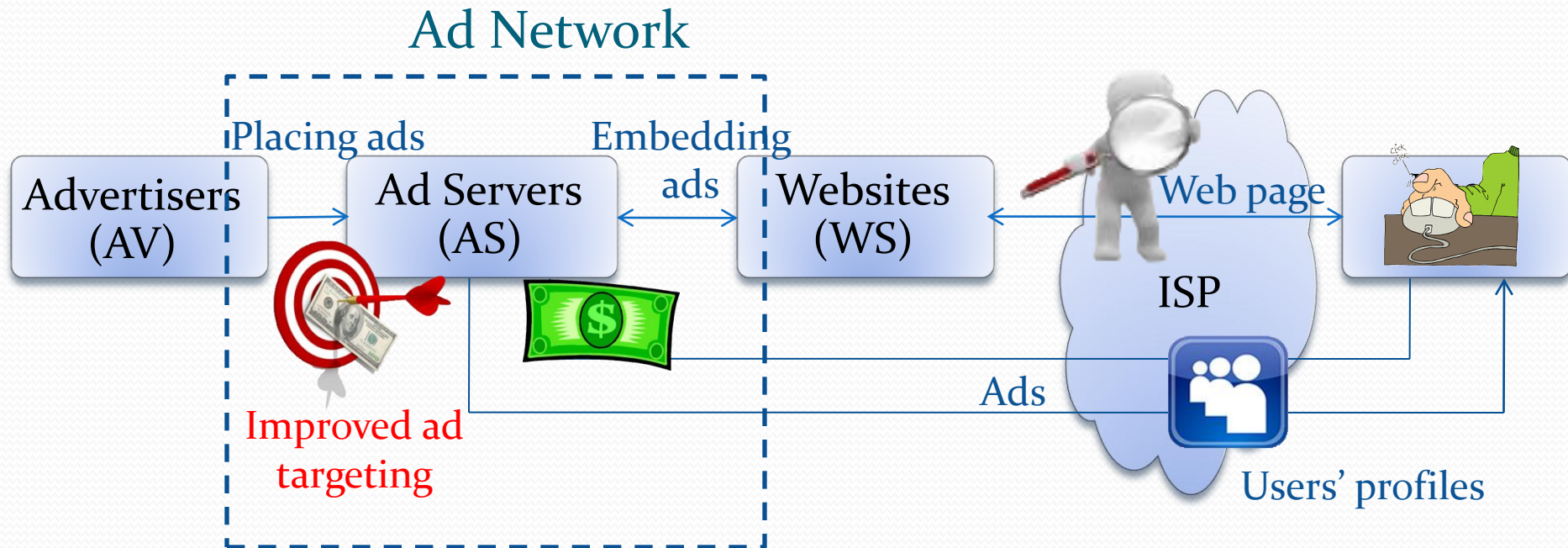
Nominal Mode

- **ISP: Abstain (A)** – forwards users' communication
- **AS: Abstain (A)** – serves online ads upon users' requests



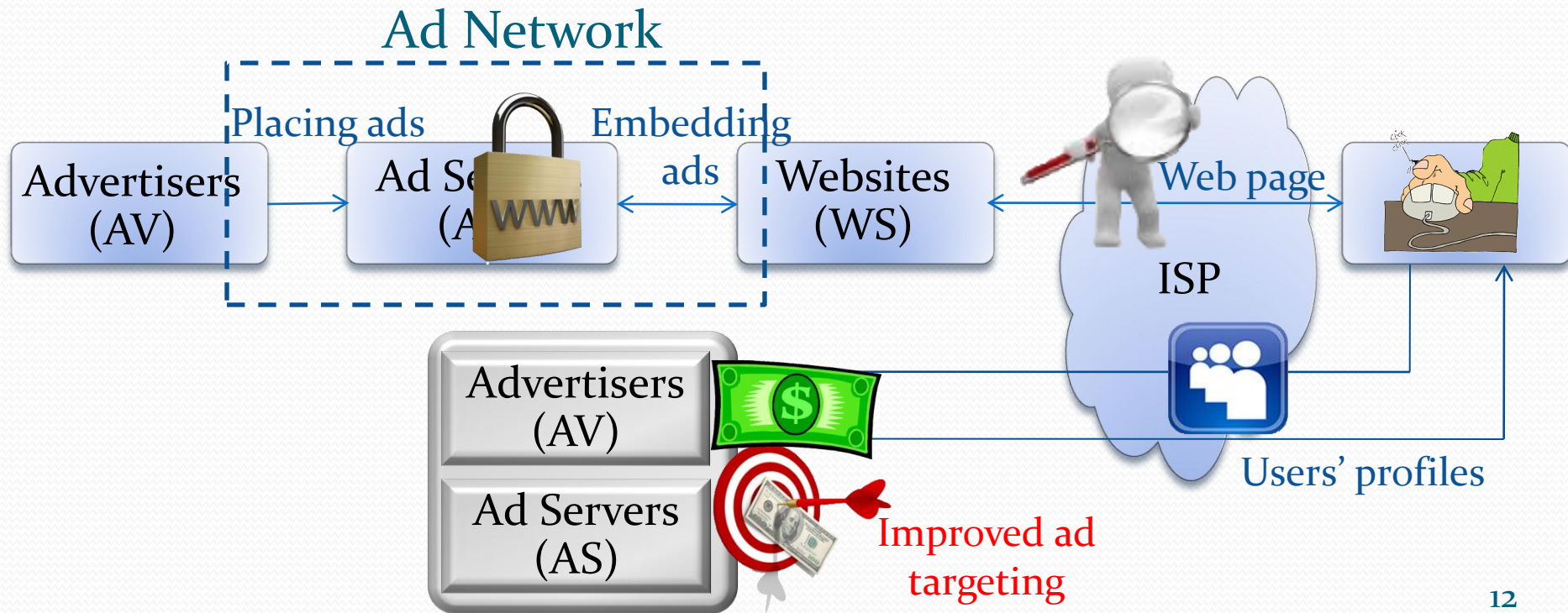
Cooperative Mode

- **ISP: Cooperate (C)** – shares the collected users' profiles to help AS improve ad targeting
- **AS: Cooperate (C)** – shares a part of its revenue with the ISP



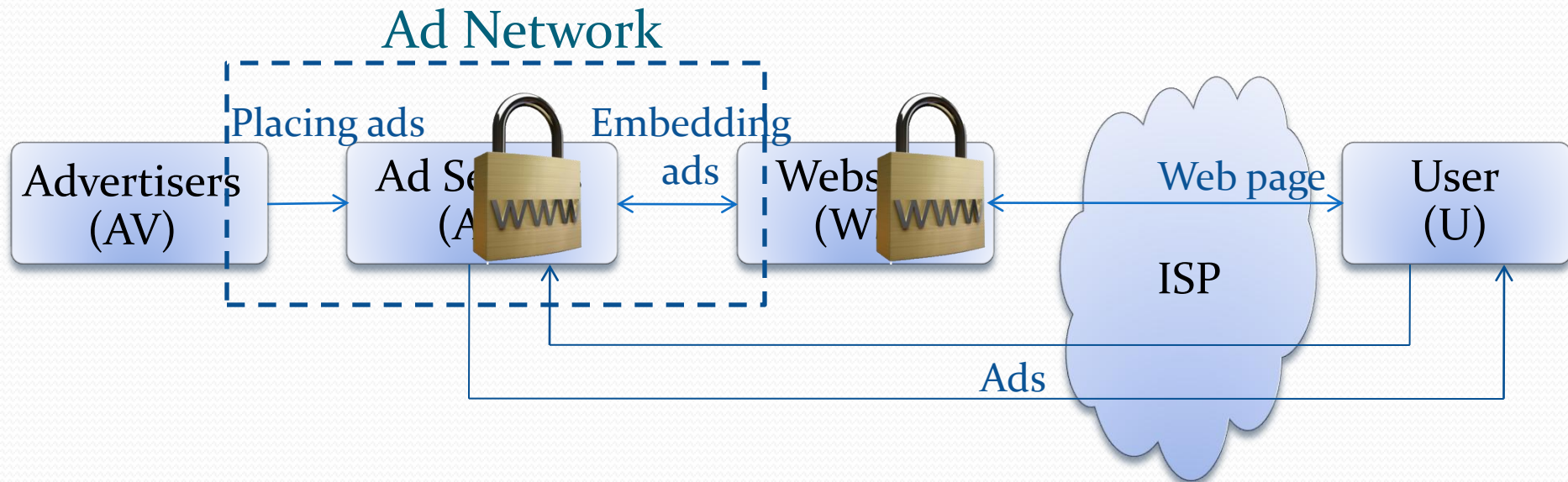
Non-Cooperative Mode

- **ISP: Divert (D)** – diverts a fraction of the ad revenue from the AS
- **AS: Abstain (A)** – serves online ads upon users' requests
- **Secure (S)** – secures the website



Non-Cooperative Mode

- **ISP: Divert (D)** – diverts a fraction of the ad revenue from the AS
- **AS: Secure (S)** – secures the website



Game-theoretic Model

- Behavior of **ISPs**:
 - **Abstain (A)** – forwards users' communication
 - **Cooperate (C)** – shares the collected users' private info to help improve ad targeting
 - **Divert (D)** – diverts a fraction of ad revenue from the AS
- Behavior of **Ad Servers (AS)**:
 - **Abstain (A)** – serves online ads upon users' requests
 - **Cooperate (C)** – shares a part of its revenue with the ISP
 - **Secure (S)** – secures a website to prevent loss of ad revenue

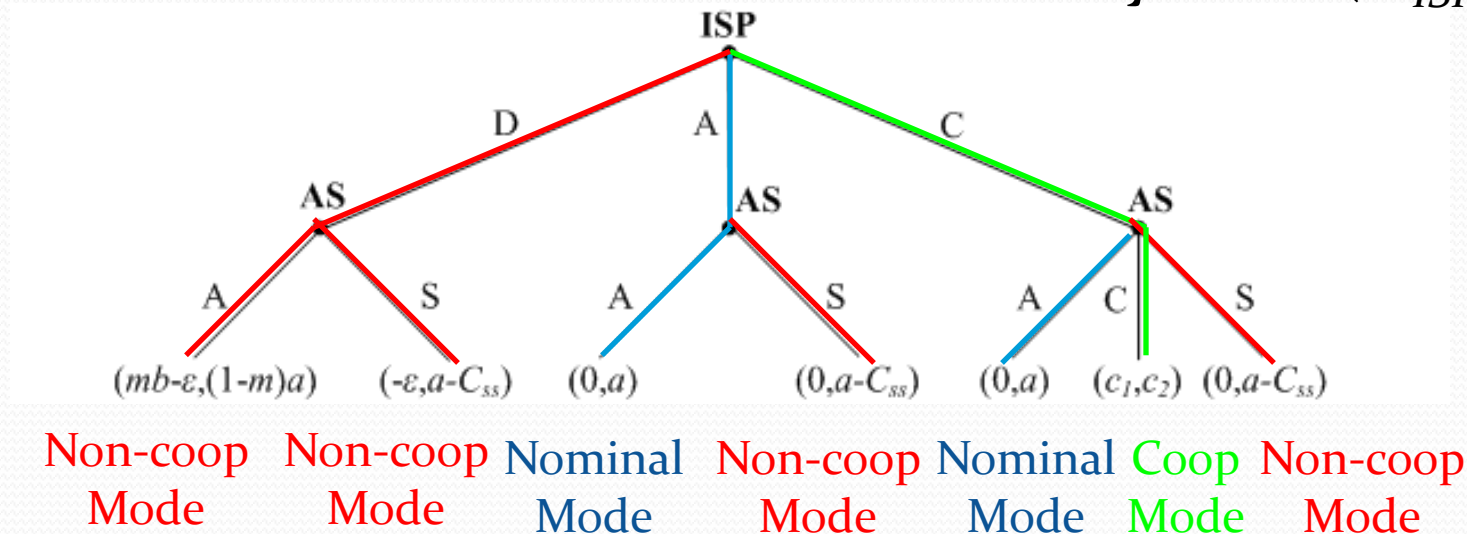
The Game

- Dynamic, finite **multi-stage** game $G=\{P, S_A, U\}$
- Set of players: $P=\{ISP, AS\}$
- Multi-stage game: *Single stage game* played for n stages
- Total payoffs over n stages= $\Sigma(\text{payoffs at each stage})$
- Complete and perfect information
- Game is modeled for a single website
- Identify Subgame Perfect Nash Equilibrium (SPNE)

Single Stage Game

- If a website is **not secured**

$$\text{Payoffs} = (U_{ISP}, U_{AS})$$

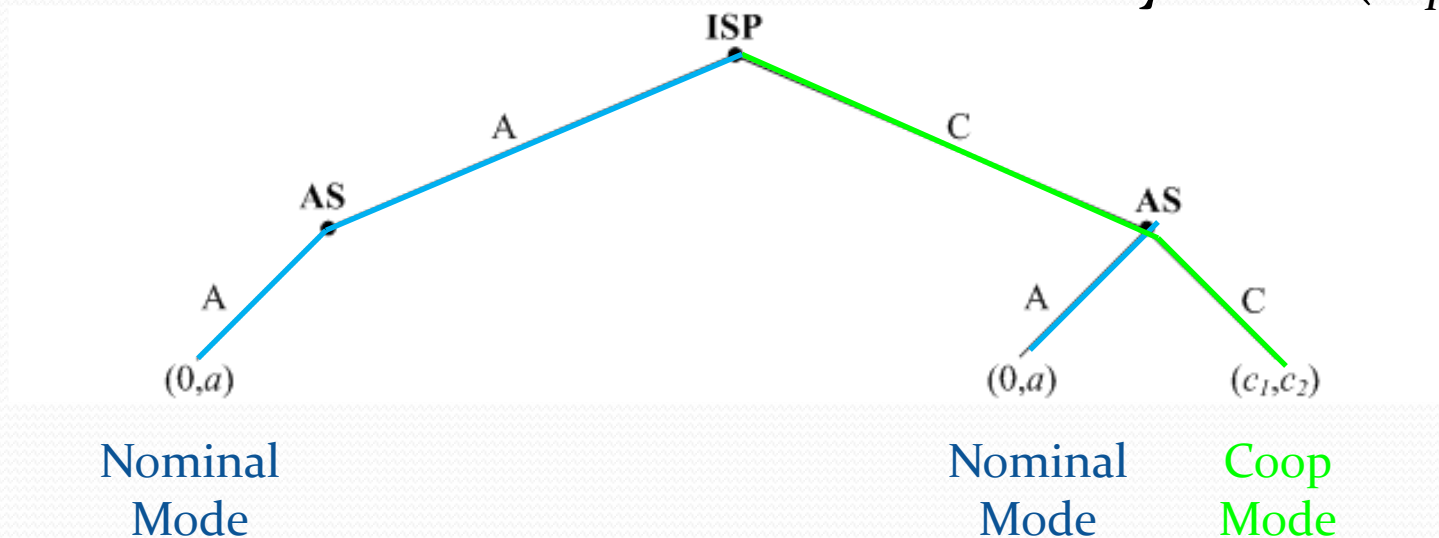


- a – AS's total payoff in the nominal mode
- c_1, c_2 – ISP's and AS's total payoff in the coop mode
- m – Fraction of clicks ISP diverts
- ε – Cost of diverting clicks
- b – ISP's per fraction revenue when diverting clicks
- C_{ss} – One-time cost of securing a website

Single Stage Game (cont'd)

- If a website is **secured**

$$\text{Payoffs} = (U_{ISP}, U_{AS})$$



- a – AS's total payoff in the nominal mode
- c_1, c_2 – ISP's and AS's total payoff in the coop mode
- m – Fraction of clicks ISP diverts
- b – ISP's per fraction revenue when diverting clicks
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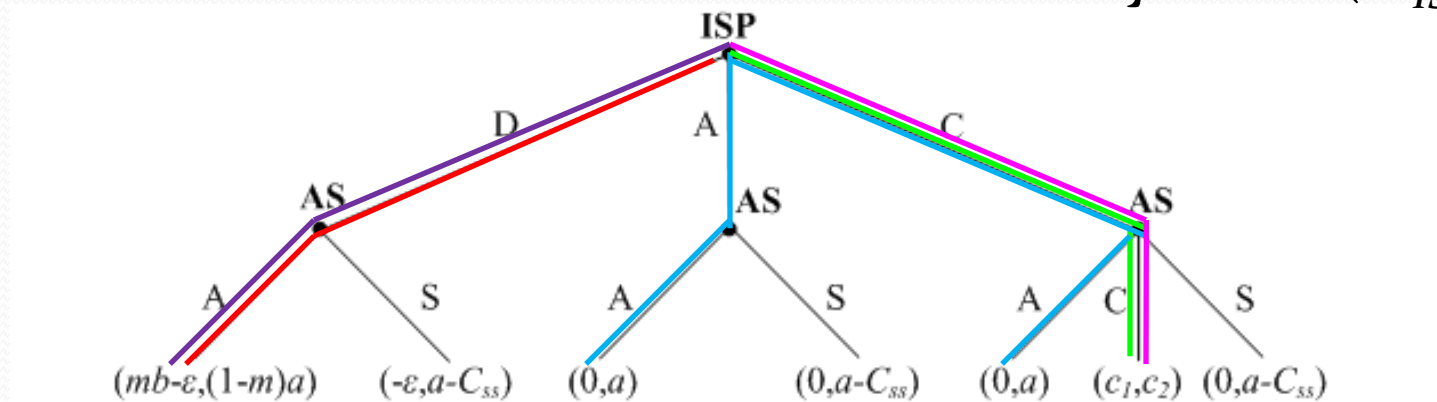
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Solving the Game

- Example: $n=1$

Payoffs = (U_{ISP}, U_{AS})



Case 1: $ma \geq C_{ss}$, $c_2 > a$

Case 2: $ma \geq C_{ss}$, $c_2 \leq a$

Case 3: $ma < C_{ss}$, $c_2 \leq a$

Case 4: $ma < C_{ss}$, $c_2 > a$, $c_1 \geq mb - \varepsilon$

Case 5: $ma < C_{ss}$, $c_2 > a$, $c_1 < mb - \varepsilon$

outcome: (C, C)

outcome: $(A, A), (C, A)$

outcome: (D, A)

outcome: (C, C)

outcome: (D, A)

Evaluations on a Real Data Set

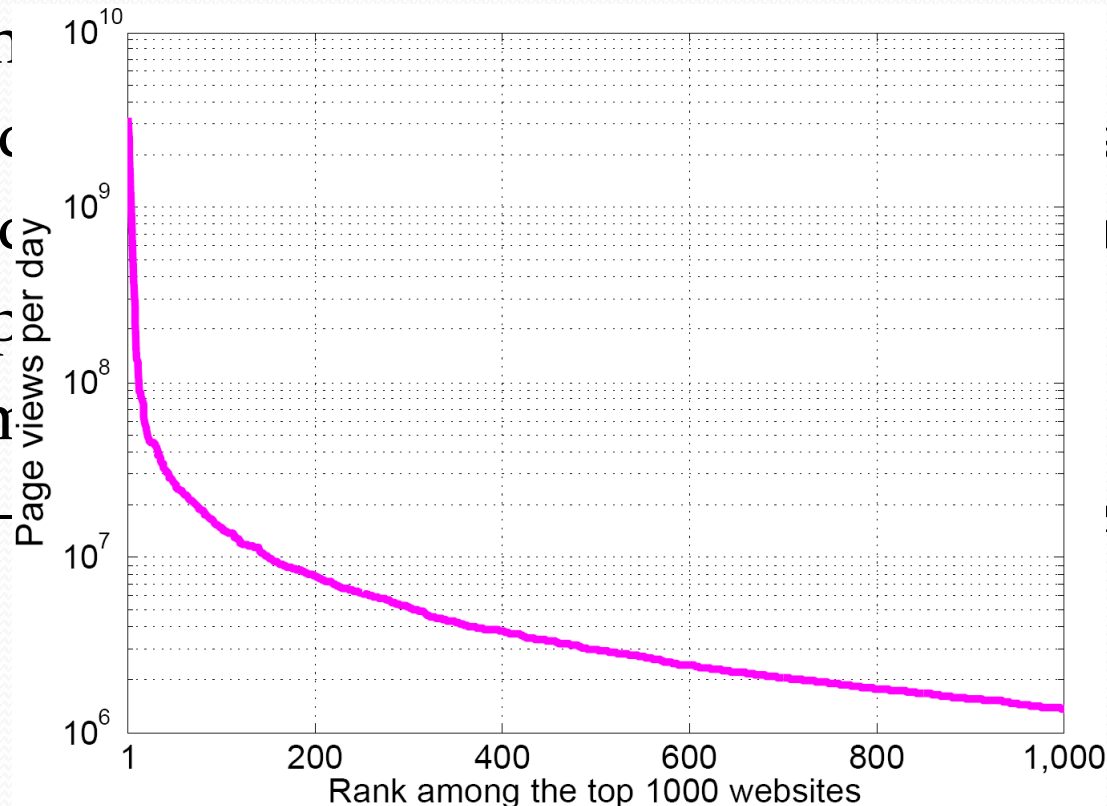
- Top 1000 most popular websites in June 2009
 - based on the data of page views [*Compete.com*]

• Parameters

- Frac
- Frac
- Imp

• Assumptions

- C_{ss} - the

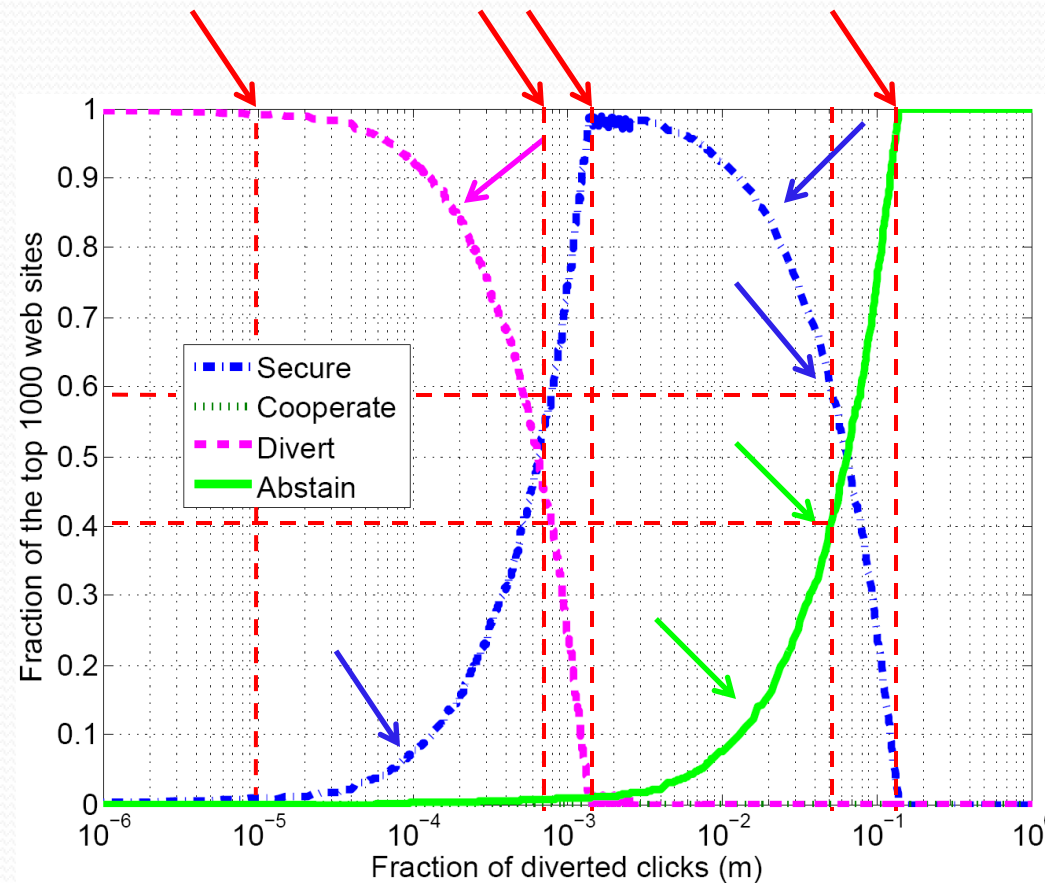


relative ISP (m)

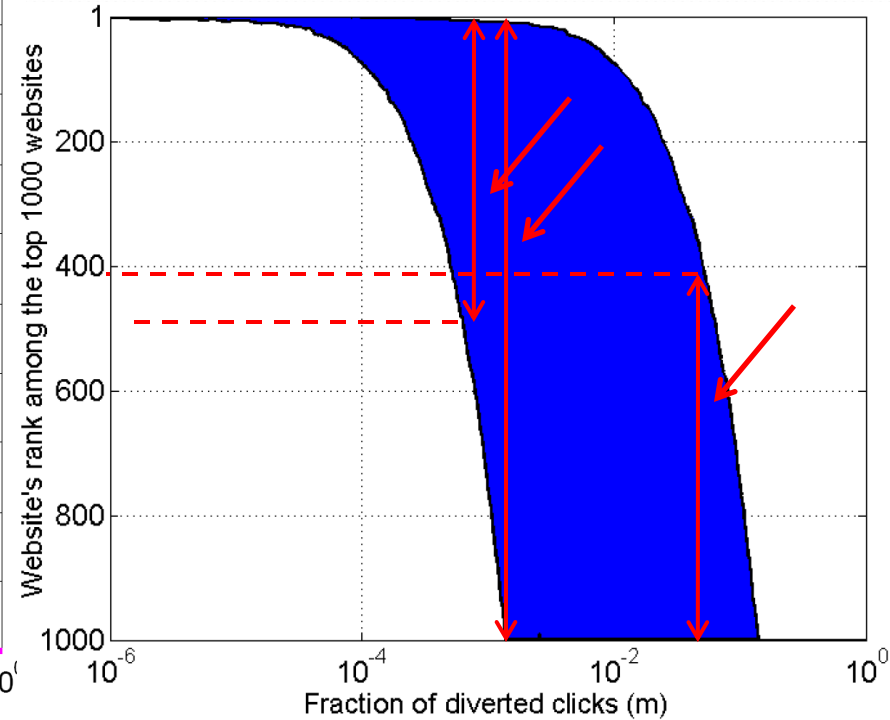
log (l)

and HTTPS at

Non-cooperative Scenario



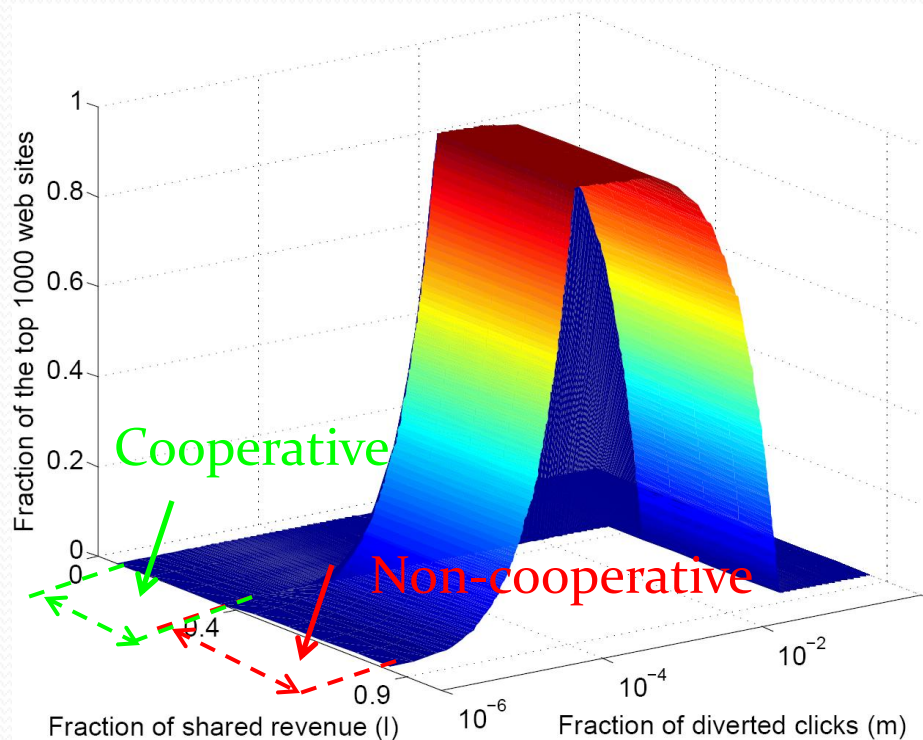
Outcomes of the multi-stage game for the top 1000 websites



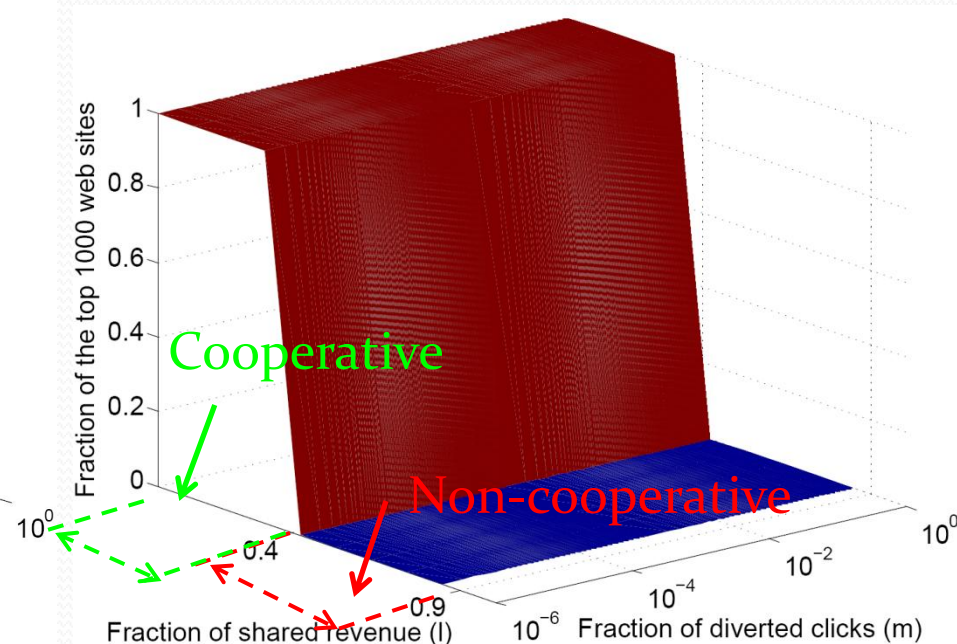
Secured websites
(secure if $ma > C_{ss}$)

Effect of the Parameters

- Fraction of shared revenue when cooperating (l)



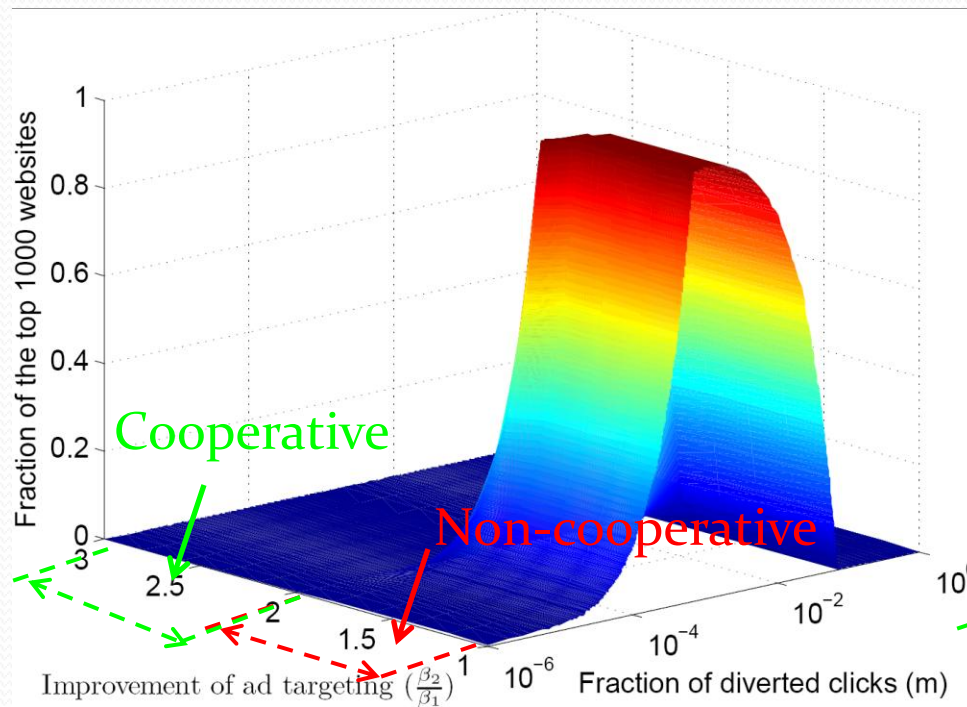
Secured websites



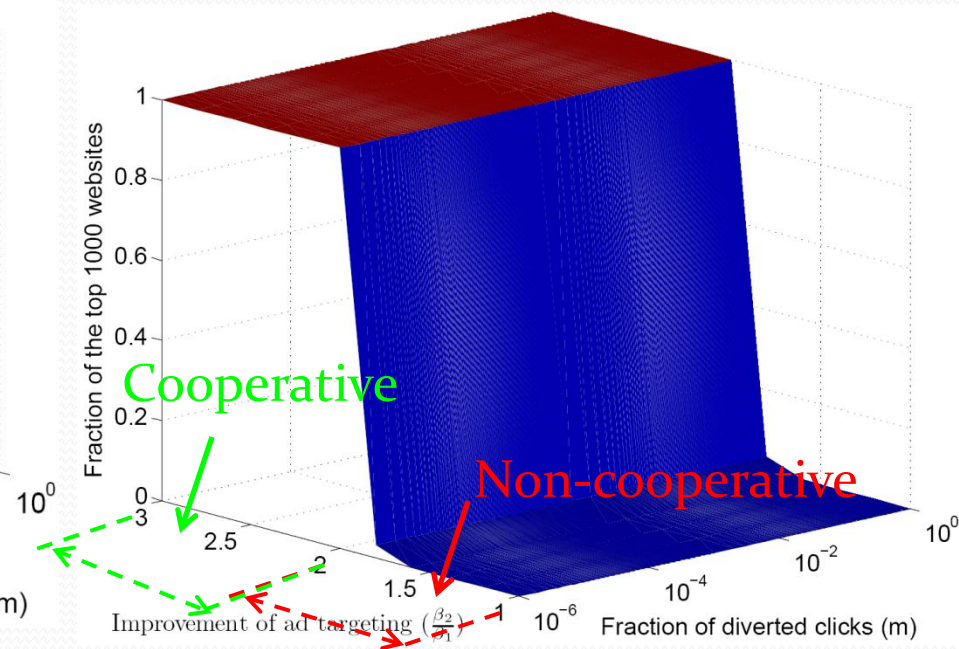
Cooperation achieved

Effect of the Parameters (cont'd)

- Improvement of ad targeting (β_2/β_1)



Secured websites



Cooperation achieved

Conclusion

- Novel problem of *ISPs becoming strategic participants* in the online advertising business
- Studied the behavior and interactions of the ISPs and ad networks
- Applied game-theoretic model to the real data
- Effect on the Web is *positive in both cases*:
 - *Cooperative ISPs*: - users receive better targeted ads
- ISPs and ad networks earn more
 - *Non-cooperative ISPs*: - improved Web security
- the most important websites secured first